



Lecture 19

A/B Testing

Announcements

- **Homework 7** due Wednesday 3/6
 - **Lab 6** due today
 - **Midterm** is 3/8 at 7pm (one week!)
 - See Ed for Midterm Prep Post
-

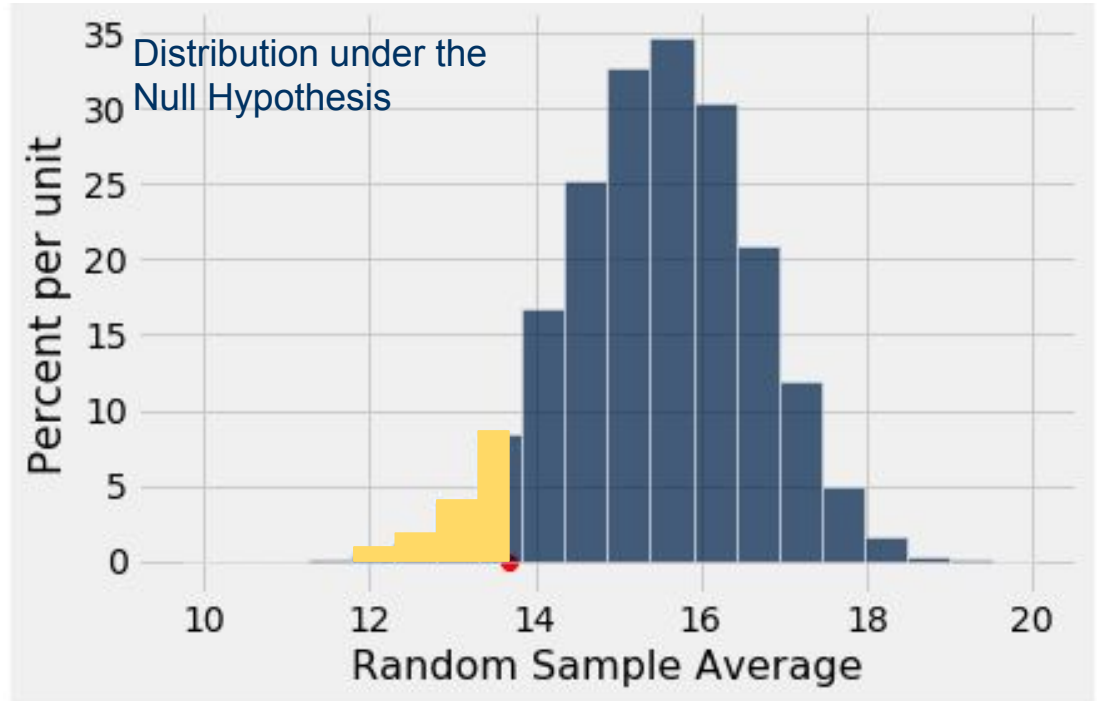
Weekly Goals

- Monday
 - Comparing distributions
 - Hypothesis tests
 - Wednesday
 - Making decisions when visualizations don't suffice
 - Tail areas and p-values
 - **Today**
 - A/B testing
 - Permutation Test
-

P-Value Review

The p-Value as an Area

- Empirical distribution of the test statistic **under the null hypothesis**.
- Red dot denotes the observed statistic.
- Yellow area denotes the tail probability (p-value).



Definition of the p -value

Formal name: **observed significance level**

The p -value is the chance (probability),

- under the null hypothesis,
 - that the test statistic
 - is equal to the value that was observed in the data
 - or is even further in the direction of the alternative.
-

A/B Testing

Comparing Two Samples

- Compare values of sampled individuals in Group A with values of sampled individuals in Group B.
- Question: Do the two sets of values come from the same underlying distribution?
- Answering this question by performing a statistical test is called **A/B testing**.

(Demo)

The Groups and the Question

- Random sample of mothers of newborns. Compare:
 - (A) Birth weights of babies of mothers who smoked during pregnancy
 - (B) Birth weights of babies of mothers who didn't smoke
 - Question: Could the difference be due to chance alone?
-

Hypotheses

- Null:
 - In the population, the distributions of the birth weights of the babies in the two groups are the same. (They are different in the sample just due to chance.)
 - Alternative:
 - In the population, the babies of the mothers who smoked weigh less, on average, than the babies of the non-smokers.
-

Test Statistic

- Group A: non-smokers
 - Group B: smokers
 - Statistic: Difference between average weights
Group B average - Group A average
 - Negative values of this statistic favor the alternative
- (Demo)
-

The Data



Non-smoker

120 oz



Non-smoker

113 oz



Smoker

128 oz



Smoker

108 oz

...



Non-smoker

...

117 oz

Shuffling Labels Under the Null



Smoker

120 oz



Non-smoker

113 oz



Non-smoker

128 oz



Smoker

108 oz

...



Smoker

117 oz

...

Shuffling Rows

Random Permutation

- `tbl.sample(n)`
 - Table of `n` rows picked randomly with replacement
- `tbl.sample()`
 - Table with same number of rows as original `tbl`, picked randomly with replacement
- `tbl.sample(n, with_replacement = False)`
 - Table of `n` rows picked randomly without replacement
- `tbl.sample(with_replacement = False)`
 - All rows of `tbl`, in random order

(Demo)

Simulating Under the Null

- If the null is true, all rearrangements of labels are equally likely
- Plan:
 - Shuffle all group labels
 - Assign each shuffled label to a birth weight
 - Find the difference between the averages of the two shuffled groups
 - Repeat

(Demo)

How We've Tested Thus Far

Hypothesis Testing Review

- **1 Sample: One Category** (e.g. percent of flowers that are purple)
 - Test Statistic: `observed_proportion, abs(observed_proportion - null_proportion)`
 - How to Simulate: `sample_proportions(n, null_dist)`
- **1 Sample: More Than 2 Categories** (e.g. ethnicity distribution of jury panel)
 - Test Statistic: `tvd(observed_dist, null_dist)`
 - How to Simulate: `sample_proportions(n, null_dist)`
- **1 Sample: Numerical Data** (e.g. scores in a lab section)
 - Test Statistic: `observed_mean, abs(observed_mean - null_mean)`
 - How to Simulate: `population_data.sample(n, with_replacement=False)`
- **2 Samples: Underlying Values** (e.g. birth weights of smokers vs. non-smokers)
 - Test Statistic: `group_a_mean - group_b_mean, group_b_mean - group_a_mean, abs(group_a_mean - group_b_mean)`
 - How to Simulate: `observed_data.sample(with_replacement=False)`